

Exam Date & Time: 10-Jan-2024 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY 2024
SUBJECT: ECE 2121- ANALOG ELECTRONIC CIRCUITS

Answer ALL questions.
Missing data may be suitably assumed.

ANALOG ELECTRONIC CIRCUITS [ECE 2121]

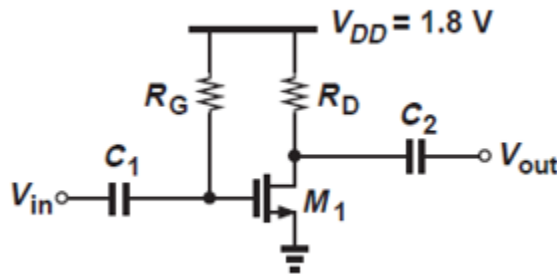
Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Design the CS stage shown in FIG.Q(1A). for a voltage gain of 5 and an output impedance of 1 k Ω . Bias the transistor so that it operates 100 mV away from the triode region. Assume the capacitors are very large and $R_D = 10$ k Ω . Assume $\mu_n c_{ox} = 200 \mu A/V^2$ & $V_{th} = 0.4V$.



(4)

FIG.Q(1A)

- 1B) Consider the CS stage shown in FIG. Q(1B)., where I_1 defines the bias current of M_1 and C_1 is very large. (a) If $\lambda = 0$ and $I_1 = 1$ mA, what is the maximum allowable value of R_D for M_1 to remain in saturation? (b) With the value found in (a), determine W/L to obtain a voltage gain of 5. Assume $\mu_n c_{ox} = 200 \mu A/V^2$ & $V_{th} = 0.4V$. (3)

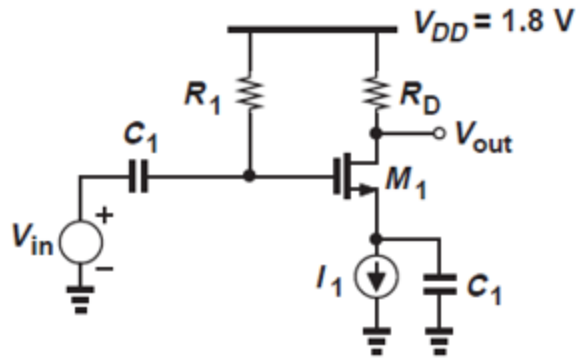
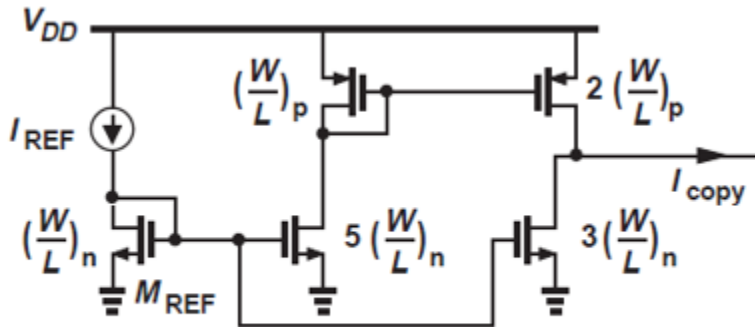


FIG.Q(1B)

- 1C) Calculate I_{copy} for the circuit shown in FIG. Q(1C). Assume all the transistors operate in saturation.



(3)

FIG.Q(1C)

- 2A) Design a common-gate stage shown in FIG.Q(2A) for the following specifications: Voltage Gain= 5, $R_S = 0$, $R_3 = 500\Omega$, $R_{in} = 50\Omega$, power budget = 2 mW, $V_{DD} = 1.8\text{ V}$. Assume $\mu_n C_{ox} = 100\text{ }\mu\text{A/V}^2$, $V_{TH} = 0.5\text{ V}$ and $\lambda = 0$.

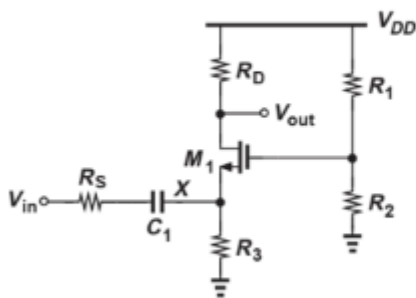


FIG.Q(2A)

(4)

- 2B) Determine the single-ended differential and common mode voltage gains for the circuit shown in FIG.Q(2B) using half circuit analysis. Assume $\lambda \neq 0$ for all the MOSFETs.

(3)

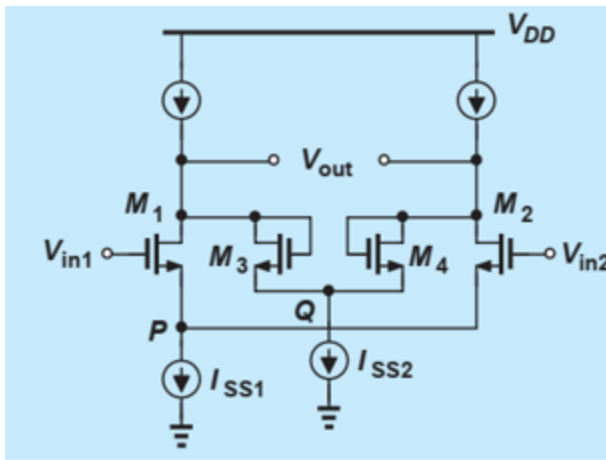


FIG.Q(2B)

- 2C) Design an NMOS differential pair for a voltage gain of 5 and a power budget of 2 mW subject to the condition that the stage following the differential pair requires an input common mode voltage level of at least 1.6 V. Assume $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $\lambda = 0$ and $V_{DD} = 1.8 \text{ V}$ (3)

- 3A) Using Miller's theorem, estimate the input capacitance of the circuit depicted in FIG. Q(3A). Assume $\lambda > 0$ but neglect other capacitances. What happens if $\lambda \rightarrow 0$?

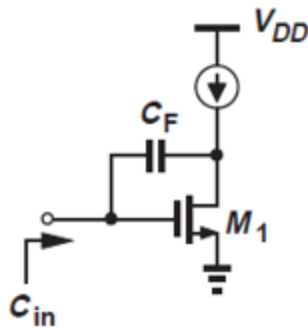


FIG.Q(3A)

- 3B) For the circuit shown in FIG.Q3(B), find the poles at low frequencies. Also, sketch its low frequency response. Assume $\lambda = 0$.

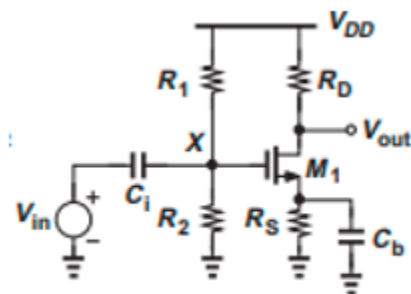


FIG.Q3(B)

- 3C) Determine the poles of the circuit shown in FIG. Q(3C). Assume $\lambda = 0$. (3)

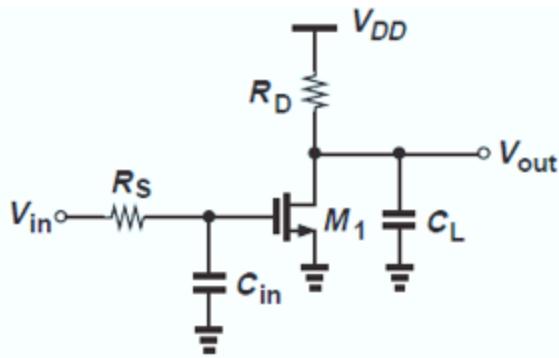


FIG. Q(3C)

- 4A) Determine the closed loop gain, I/O impedances of the circuit shown in FIG.Q(4A) by considering I/O effects. Assume $\lambda=0$ for both M_1 and M_2 .

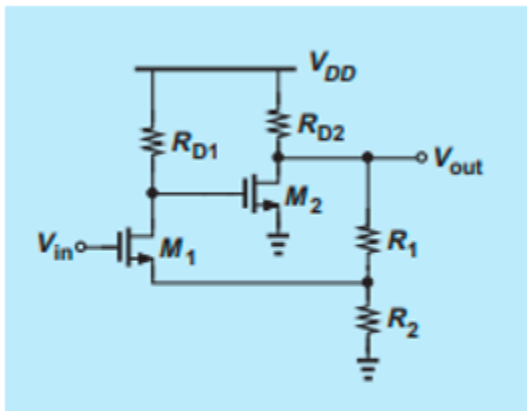


FIG. Q(4A)

- 4B) Determine the polarity of feedback for the circuits shown in FIG.Q4B(i) and FIGQ4B(ii). Explain the steps involved in the analysis.

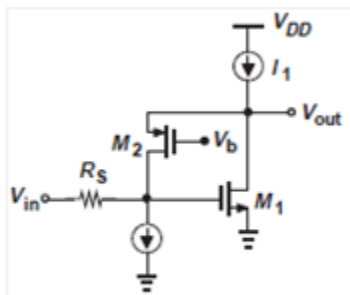


FIG. Q4B(i)

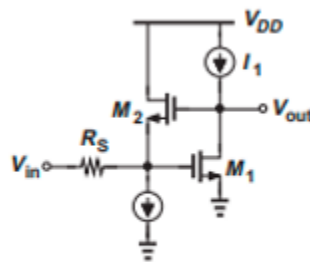


FIG. Q4B(ii)

- 4C) Show that the negative feedback in amplifier improves the bandwidth and stability of the system. (3)
- 5A) For the circuit shown in FIG.Q5(A), (4)
- Obtain an expression for the transfer function of each stage.
 - Write the expression for the overall transfer function.
 - Obtain an expression for the oscillation frequency

iv) If $R_D = 2\text{k}\Omega$, and $C_D = 0.1\mu\text{F}$, at what frequency the circuit will oscillate?

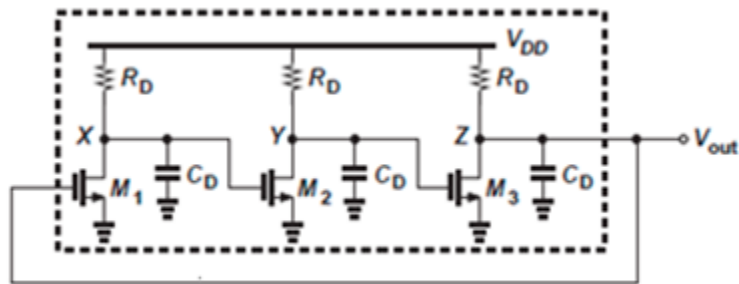


FIG. Q5(A)

- 5B) Explain the working principle of Wein Bridge oscillator with a neat circuit diagram. Obtain the expressions for minimum gain and frequency of oscillation. (3)
- 5C) Obtain an expression for the efficiency of Class - B push pull power amplifier. (3)

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